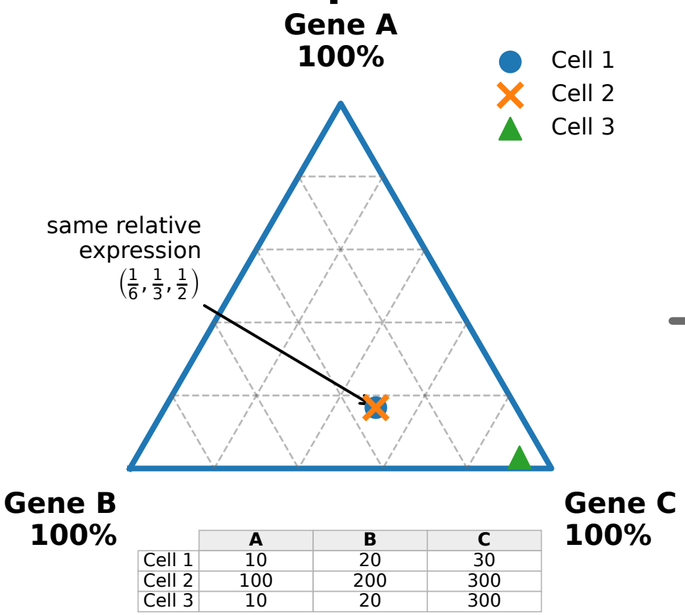
clrcycle workflow

1. Composition



	A	B	C
Cell 1	10	20	30
Cell 2	100	200	300
Cell 3	10	20	300

2. CLR coordinates

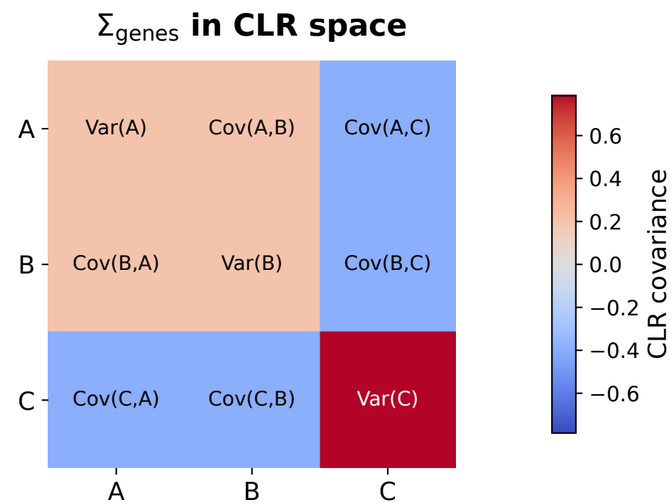
$$z_{ig} = \log \frac{x_{ig}}{g(\mathbf{x}_i)}$$

gene expression x_{ig} relative to the cell geometric mean $g(\mathbf{x}_i)$

$$\mathrm{clr}(c\mathbf{x}) = \mathrm{clr}(\mathbf{x})$$

scale invariant

3. Gene covariance



4. Optimize gene order

$$F(\pi) = \mathbf{c}^T \Sigma_\pi \mathbf{c} + \mathbf{s}^T \Sigma_\pi \mathbf{s}$$

π: candidate gene order Σ_π: CLR covariance reordered by π

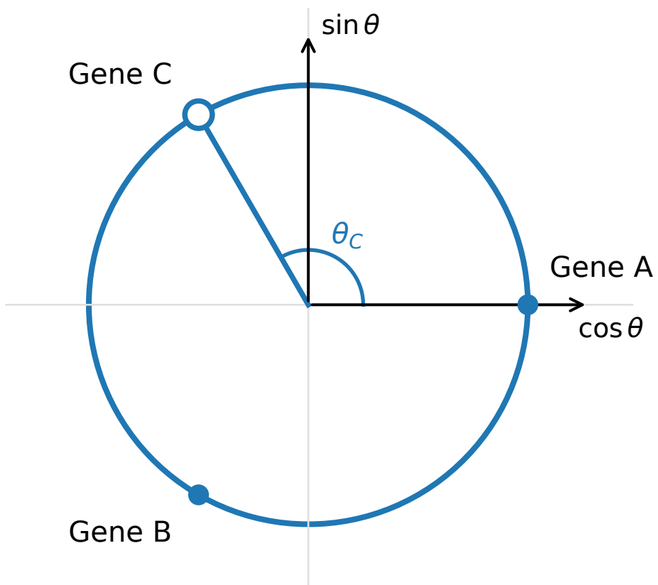
$$\theta_g = 2\pi(g-1)/G$$

$$c_g = \sqrt{\tfrac{2}{G}} \cos\theta_g, \qquad s_g = \sqrt{\tfrac{2}{G}} \sin\theta_g$$

$$\hat{\pi} = \arg\max_{\pi} F(\pi)$$

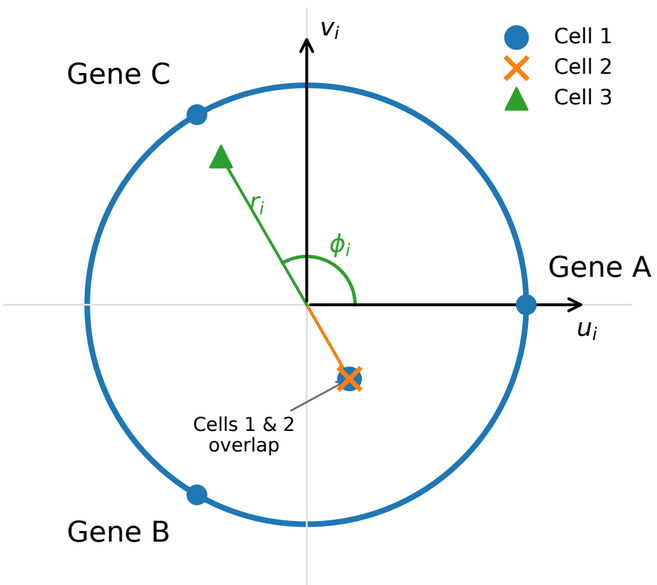
$\hat{\pi}$ is the optimal gene order

5. First Fourier mode



$$\theta_C = \frac{2\pi(2-1)}{3} = \frac{2\pi}{3}$$

6. Project cells



$$u_i = \sum_g Z_{i,\hat{\pi}(g)} c_g \qquad v_i = \sum_g Z_{i,\hat{\pi}(g)} s_g$$